

Original practice 2: Networks and branch evidence

Original prediction practice—not official questions or a guarantee

Total: 36 marks • Suggested time: 75 minutes

Series/parallel topology, inverse network design, branch faults and current/voltage consistency.

Status and instructions

Original prediction practice—not official questions or a guarantee

Author-created chapter practice, not a claim about an official examination format. All 16 questions are compulsory. Show working and units. Circuit wires, meters and sources are ideal unless stated; resistor temperatures remain constant unless a question explicitly studies a change. Treat stated data as exact unless a tolerance is given.

All case datasets and SVGs are original constructed teaching material, not measurements reported by NCERT. NCERT is cited for concepts and formulas, not as the source of these questions.

- A: Q1–6, $6 \times 1 = 6$ marks
- B: Q7–10, $4 \times 2 = 8$ marks
- C: Q11–13, $3 \times 3 = 9$ marks
- D: Q14, $1 \times 5 = 5$ marks
- E: Q15–16, $2 \times 4 = 8$ marks
- Total: 16 questions, 36 marks

Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026–27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm’s law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers V–I measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming “Electric Effects of Electric Current” among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV’s 13 marks covers a combined unit, not an Electricity-only allotment.

Section A — MCQ

1. [1 marks]

1. Two branches between the same nodes contain $8\ \Omega$ and $24\ \Omega$. What is their equivalent resistance?

- A. $6\ \Omega$
- B. $16\ \Omega$
- C. $32\ \Omega$
- D. $192\ \Omega$

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.2, printed pp185–187; Activity 11.6 pp185–186; Eqs. (11.15)–(11.18) p186: branch-current sum and reciprocal-resistance sum. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

2. [1 marks]

2. A $4\ \Omega$ resistor and an $8\ \Omega$ resistor are in series. Their potential-difference magnitudes have which ratio $V_4:V_8$?

- A. 2:1
- B. 1:1
- C. 4:1
- D. 1:2

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3. [1 marks]

3. A third branch with positive, nonzero, finite resistance is added in parallel across an ideal source maintaining a fixed nonzero voltage. What happens to the magnitude of total source current?

- A. Decreases
- B. Increases
- C. Becomes zero
- D. Must remain unchanged

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4. [1 marks]

4. In a steady junction, 1.2 A arrives and splits into two branches. One branch carries 0.45 A. What does the other carry?

- A. 1.65 A
- B. 0.54 A
- C. 0.75 A
- D. 2.67 A

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5. [1 marks]

5. Two equal $10\ \Omega$ resistors are connected in parallel across 5 V. What is the current through each resistor?

- A. 0.50 A
- B. 1.0 A
- C. 0.25 A
- D. 2.0 A

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6. [1 marks]

6. A resistor is disconnected so that one of two parallel branches becomes open. With an ideal fixed-voltage source, the current in the other unchanged branch will:

- A. Double
- B. Become zero
- C. Halve
- D. Remain unchanged

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Section B — 2-mark questions

7. [2 marks]

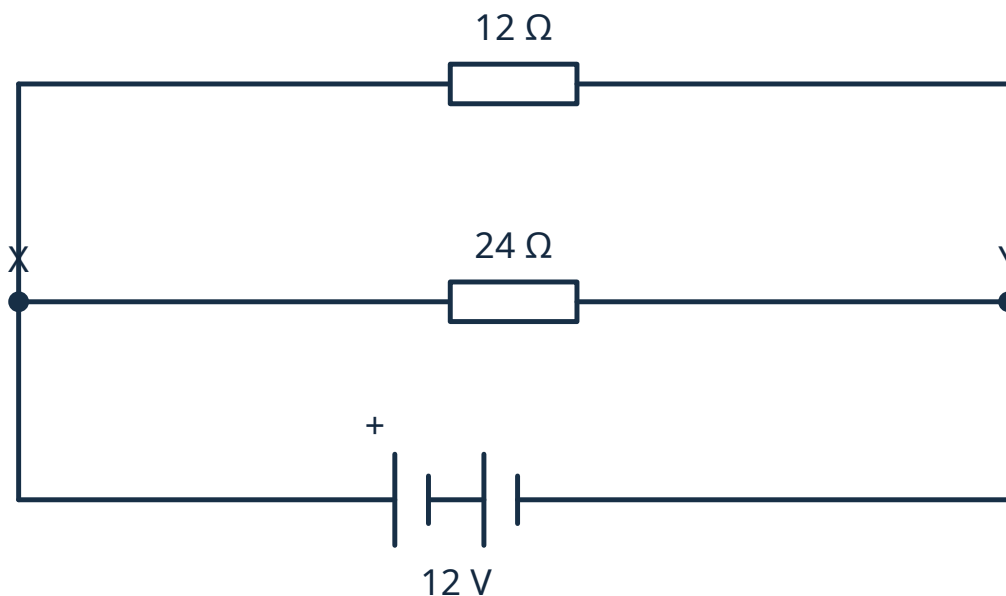
7. A $7\ \Omega$ and an unknown resistor in series draw 0.50 A from 8 V. Determine the total resistance and the unknown resistance.

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8. [2 marks]

8. Find the two branch currents in the supplied 12 V network containing $12\ \Omega$ and $24\ \Omega$.

12 V parallel circuit



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9. [2 marks]

9. A learner says that every $5\ \Omega$ resistor must dissipate the same power. Give two different voltages and the resulting powers to test this claim; assume constant resistance.

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; $1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

10. [2 marks]

10. An $18\ \Omega$ resistor is connected in parallel with an unknown resistor. Their equivalent resistance is $6\ \Omega$. Find the unknown resistance.

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Section C — 3-mark questions

11. [3 marks]

11. Two resistors, $4\ \Omega$ and $12\ \Omega$, are first joined in series and then in parallel across the same $12\ \text{V}$ source. Calculate (a) series current, (b) parallel equivalent resistance and (c) total parallel current.

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12. [3 marks]

12. You have one $5\ \Omega$ resistor and two $10\ \Omega$ resistors. Use all three to obtain $10\ \Omega$. Describe the connections, calculate the equivalent resistance and find source current at $15\ \text{V}$.

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13. [3 marks]

13. In a parallel pair, one branch carries $0.30\ \text{A}$ and the other $0.20\ \text{A}$. The source is $6\ \text{V}$. Determine (a) both branch resistances (1 mark together), (b) total current (1) and (c) equivalent resistance (1).

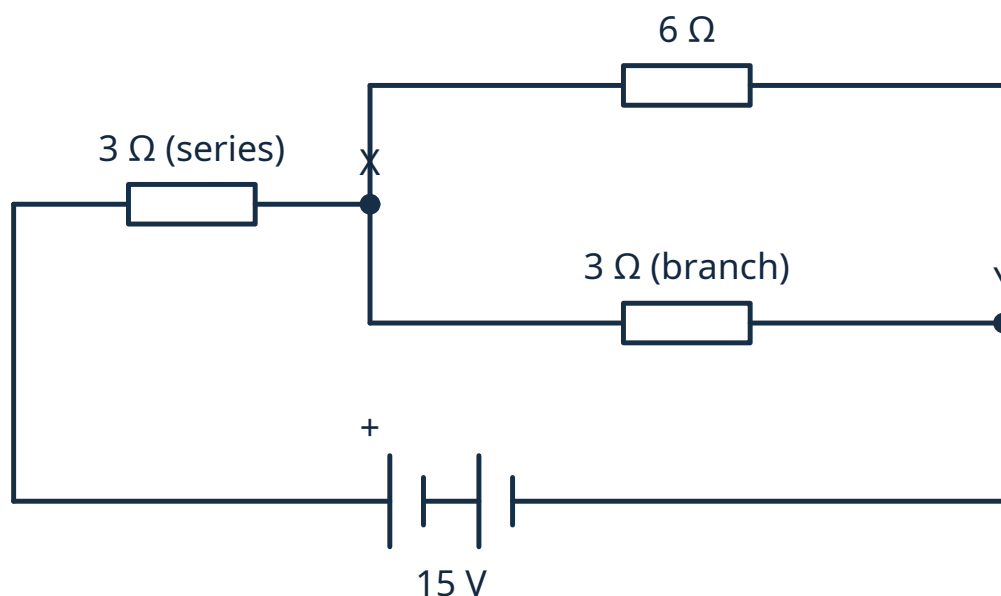
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Section D — 5-mark questions

14. [5 marks]

14. In the supplied 15 V circuit, a $3\ \Omega$ series resistor feeds a parallel pair of $6\ \Omega$ and $3\ \Omega$. Find (a) parallel equivalent resistance, (b) total resistance, (c) source current, (d) voltage across the parallel pair and (e) current through its $6\ \Omega$ branch. Each part is worth 1 mark.

15 V series-parallel circuit



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Section E — Case-based questions

15. [4 marks]

15. Constructed case data for an ideal 8 V supply: parallel branches P and Q carry 0.40 A and 0.20 A. On opening Q, a second sheet records source current = 0.40 A, P current = 0.40 A, P voltage = 8 V. (a) Find R_P (1). (b) Find R_Q before opening (1). (c) Find total current before opening (1). (d) Explain whether the second sheet is consistent (1).

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.2, printed pp185–187; Activity 11.6 pp185–186; Eqs. (11.15)–(11.18) p186: branch-current sum and reciprocal-resistance sum. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

16. [4 marks]

16. Constructed test data: a sealed box contains two positive resistors, each larger than zero, and no other components. At a controlled temperature, readings are $(V, I) = (3, 0.5), (6, 1.0), (9, 1.5)$, in V and A. The resistors are labelled $9\ \Omega$ and $18\ \Omega$. (a) Infer the box resistance from the data (1). (b) Calculate the series value (1). (c) Calculate the parallel value (1). (d) Identify which of these two arrangements fits, and explain why terminal data alone would not identify an arbitrary unknown network (1).

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Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — original-practice-02-q01

[1 mark] A. $1/R = 1/8 + 1/24 = 4/24$, so $R = 6 \Omega$.

Question 2 — original-practice-02-q02

[1 mark] D. Same current gives $V_4/V_8 = 4/8 = 1/2$, so the ratio is 1:2.

Question 3 — original-practice-02-q03

[1 mark] B. The new branch draws additional current while existing branch currents remain unchanged, so total current increases.

Question 4 — original-practice-02-q04

[1 mark] C. $I_{\text{other}} = 1.2 - 0.45 = 0.75 \text{ A}$.

Question 5 — original-practice-02-q05

[1 mark] A. Each has 5 V, so each current is $5/10 = 0.50 \text{ A}$.

Question 6 — original-practice-02-q06

[1 mark] D. Its voltage and resistance remain the same, so its current is unchanged.

Question 7 — original-practice-02-q07

[1 mark] $R_{\text{total}} = V/I = 8/0.50 = 16 \Omega$.

[1 mark] $R_{\text{unknown}} = 16 - 7 = 9 \Omega$.

Question 8 — original-practice-02-q08

[1 mark] Both branches have 12 V. $I_{12} = 12/12 = 1.0 \text{ A}$.

[1 mark] $I_{24} = 12/24 = 0.50 \text{ A}$.

Question 9 — original-practice-02-q09

[1 mark] For example, at 5 V: $P = V^2/R = 25/5 = 5 \text{ W}$.

[1 mark] At 10 V: $P = 100/5 = 20 \text{ W}$. Thus equal resistance alone does not imply equal power.

Question 10 — original-practice-02-q10

[1 mark] $1/6 = 1/18 + 1/R$, so $1/R = 1/6 - 1/18 = 1/9$.

[1 mark] $R = 9 \Omega$; a 6Ω equivalent is smaller than either branch, consistent with parallel connection.

Question 11 — original-practice-02-q11

[1 mark] (a) $R_s = 4 + 12 = 16 \Omega$; $I_s = 12/16 = 0.75 \text{ A}$.

[1 mark] (b) $R_p = (4 \times 12)/(4 + 12) = 3 \Omega$.

[1 mark] (c) $I_p = 12/3 = 4 \text{ A}$.

Question 12 — original-practice-02-q12

[1 mark] Connect the two $10\ \Omega$ resistors in parallel; put that pair in series with $5\ \Omega$.

[1 mark] $R_{\text{pair}} = 10 \times 10 / (10 + 10) = 5\ \Omega$; $R_{\text{total}} = 5 + 5 = 10\ \Omega$.

[1 mark] $I_{\text{total}} = 15 / 10 = 1.5\ \text{A}$.

Question 13 — original-practice-02-q13

[1 mark] (a) $R_1 = 6 / 0.30 = 20\ \Omega$ and $R_2 = 6 / 0.20 = 30\ \Omega$; award 0.5 mark each.

[1 mark] (b) $I_{\text{total}} = 0.30 + 0.20 = 0.50\ \text{A}$.

[1 mark] (c) $R_{\text{eq}} = 6 / 0.50 = 12\ \Omega$.

Question 14 — original-practice-02-q14

[1 mark] (a) $R_p = 6 \times 3 / (6 + 3) = 2\ \Omega$.

[1 mark] (b) $R_{\text{total}} = 3 + 2 = 5\ \Omega$.

[1 mark] (c) $I_{\text{total}} = 15 / 5 = 3\ \text{A}$.

[1 mark] (d) $V_{XY} = I_{\text{total}} R_p = 3 \times 2 = 6\ \text{V}$ (the series drop is $9\ \text{V}$).

[1 mark] (e) $I_6 = 6 / 6 = 1\ \text{A}$; the $3\ \Omega$ branch carries $2\ \text{A}$, so branch currents sum to $3\ \text{A}$.

Question 15 — original-practice-02-q15

[1 mark] (a) $R_P = 8 / 0.40 = 20\ \Omega$.

[1 mark] (b) $R_Q = 8 / 0.20 = 40\ \Omega$.

[1 mark] (c) $I_{\text{total}} = 0.40 + 0.20 = 0.60\ \text{A}$.

[1 mark] (d) Yes. P retains $8\ \text{V}$ and $20\ \Omega$, so it still draws $0.40\ \text{A}$; open Q draws zero and source current equals P current.

Question 16 — original-practice-02-q16

[1 mark] (a) All V/I ratios equal $6\ \Omega$.

[1 mark] (b) Series gives $9 + 18 = 27\ \Omega$, not $6\ \Omega$.

[1 mark] (c) Parallel gives $9 \times 18 / (9 + 18) = 6\ \Omega$.

[1 mark] (d) Parallel fits the stated two choices. Other networks can also have $6\ \Omega$ equivalent resistance, so the inference relies on the stated components and choices.